

Avery Fisher

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SUMMARY

Spacecraft test engineer with 6 years across space and defense hardware — vehicle-level integration test, RF and digital troubleshooting, and test procedure development. Builds Python and AI-assisted tooling that removes manual effort from test campaigns, most recently replacing a multi-month wire-tracing effort with graph-based signal-chain analysis.

EXPERIENCE

Blue Origin

July 2025 - Present

Spacecraft Test Engineer III

Huntsville, AL

- Designed and built an AI-assisted harness test-mapping tool using graph-based signal-chain analysis, auto-generating validated CIRRIS test files in minutes and replacing a multi-month manual wire-tracing effort across 23,000+ vehicle connections.
- Delivered ~\$400K in labor savings and eliminated operator transcription errors; produced a gap analysis report that de-risked integration by surfacing untested connections before mate. Now used across the program for harness test file generation.
- Developed and led the Powered Safe-to-Mate campaign for the Blue Ring vehicle — defined test requirements with responsible engineers, authored the 120+ operation procedure, and drove it through the formal release cycle to on-schedule execution, catching defects at vehicle level that box-level ATP had missed.
- Sustained a ~2-month test surge across 1st shift, 2nd shift, and weekends to protect critical program schedule, directing technicians and deploying custom software into production for the first time to re-author test cases when as-built vehicle configuration changed.
- Became the team's go-to resource for test procedure development — produced job aids, standardized data templates, and technician instructions that reduced reliance on tribal knowledge and let new engineers execute test campaigns independently.
- Authored and led test readiness reviews, presenting readiness and risk to stakeholders to secure approval to proceed.

Raytheon

December 2022 – July 2025

Electrical Engineer II

Forest, MS

- Conducted root cause analysis and troubleshooting of test failures, collaborating with design and supplier quality teams to identify and resolve issues promptly.
- Designed and implemented a Python-based QR code scanning application for module programming, successfully eliminating all serialization-related defects.
- Facilitated and led regular tech team meetings to foster effective communication and collaboration among team members, establishing a new “Yield Drivers” section that leverages yield data to prioritize engineering investigations and reduce field failures.
- Reported test statuses to operations, ensuring transparency and proactive responses to challenges.
- Provided cross-functional engineering support to an understaffed program, traveling off-site to document and refine testing/manufacturing processes and present staffing/testing recommendations to leadership.
- Led multiple high-impact engineering investigations—securing a supplier design change to prevent factory and field failures, developing an x-ray screening method that cut RF filter verification time from over an hour to 5–10 minutes, and driving a site-wide AI optical inspection initiative to reduce defects across multiple programs.

Raytheon

May 2020 – December 2022

Electrical Engineer I

Forest, MS

- Developed, maintained, and executed testing procedures to verify the performance and reliability of aircraft radar systems including RF, digital, and power supply testing.
- Diagnosed and troubleshot complex electrical systems, utilizing advanced testing equipment, electrical schematics, wiring diagrams, and testing methodologies to identify and resolve issues promptly.
- Successfully deployed a radar signal processing test station, including equipment setup, calibration, and configuration, enabling efficient and reliable testing of products.
- Provided specialized training and resources to technicians, reducing test station downtime.

AWARDS & HONORS

- Tech Honors, Raytheon (awarded to top 3% of technical performers based on peer nominations) **2024**

EDUCATION

Mississippi State University **December 2023**
Master of Science in Electrical and Computer Engineering Starkville, MS

GPA 4.00/4.00

Coursework: Computer Architecture, Digital Signal Processing, Optimization for Machine Learning, Literate Programming, Autonomous Vehicle Sensor Processing, Advanced Robotics, Design & Implementation of Wearables

Mississippi State University **May 2020**
Bachelor of Science in Electrical Engineering Starkville, MS

GPA: 3.79/4.00

Coursework: Electromagnetics I/II, Introduction to Radar, Circuits I/II/III, Signals and Systems, Microprocessors
Organizations: Tau Beta Pi (*Top 1/4 of Engineering Class*), Eta Kappa Nu (*Top 1/3 of Electrical Engineering Class*)

TECHNICAL SKILLS

- **Programming and Scripting:** Python (pandas, NumPy, NetworkX), C/C++, Assembly
- **Software & Tools:** MATLAB (data analysis & visualization), LabVIEW (instrument control & data acquisition), ScanWorks (boundary-scan automation), Git, AI-assisted development (Claude), MS Office
- **Hardware:** Oscilloscopes, Multimeters, Function Generators, Spectrum Analyzers, Network Analyzers, Signal Generators, CIRRIS cable tester